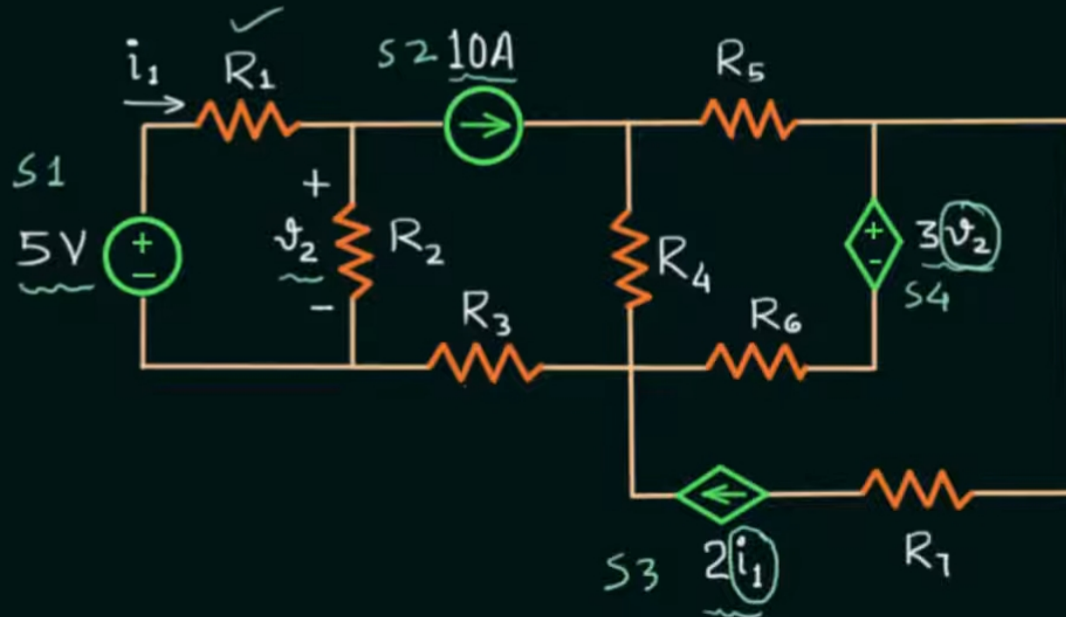


R , L and C

Dependent and Independent Sources

Independent Source: The element for which both voltage and current don't depend on the voltage or current elsewhere in the circuit.

Dependent Source: The element for which either the voltage or current depends on the voltage or current elsewhere in the circuit.



$S1 \rightarrow$ Independent

$S2 \rightarrow$ Independent

$S3 \rightarrow$ Dependent

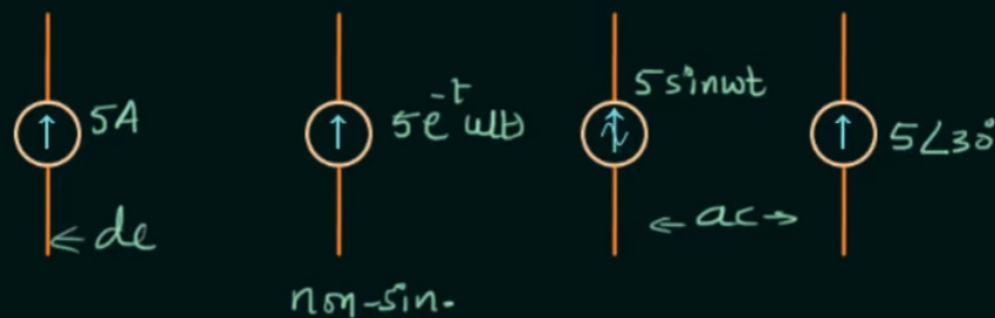
$S4 \rightarrow$ Dependent



→ Independent Voltage Source Representations:



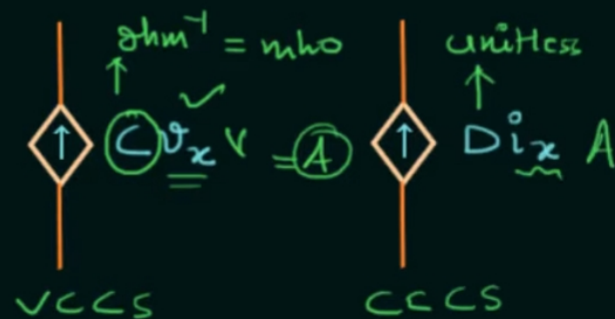
→ Independent Current Source Representations:

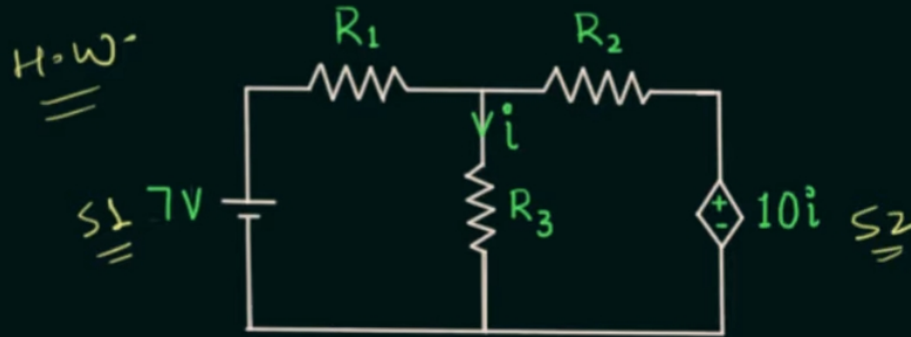


→ Dependent Voltage Source Representations:



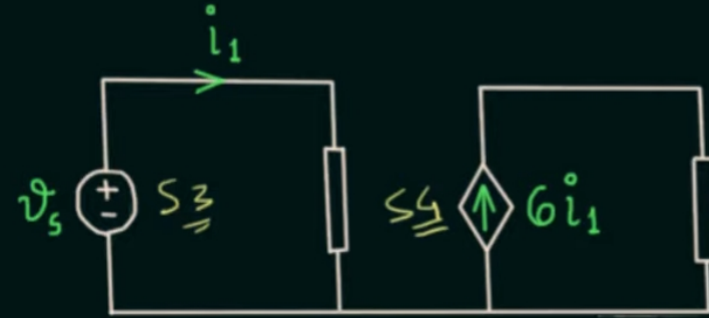
→ Dependent Current Source Representations:





S1 - Independent Voltage source

S2 - Current controlled Current Source



S3 - Independent Voltage source

S4 - Current controlled Current source